3D Vectors & the Cross Product

Vector arithmetic in three dimensions

1 Given $\mathbf{a} = (2, -1, 3)$ and $\mathbf{b} = (1, 4, -2)$, find $\mathbf{a} + 2\mathbf{b}$.

2 Find the magnitude of $\mathbf{v} = (6, -3, 2)$.

The scalar (dot) product in 3D

3 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 1, -2)$ and $\mathbf{b} = (-1, 4, 2)$, and use it to determine whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

4 Find the angle between $\mathbf{a} = (1, 0, 1)$ and $\mathbf{b} = (0, 1, 1)$, to the nearest degree.

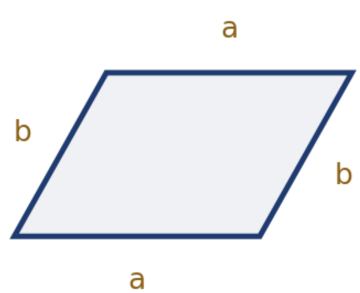
The vector (cross) product

5 Find $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = (2, 1, -1)$ and $\mathbf{b} = (1, 3, 2)$, using $\mathbf{a} \times \mathbf{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$.

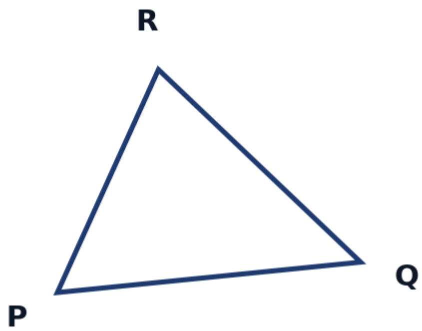
6 Explain why $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, and verify this for the result found in the previous problem by computing $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a}$.

Area of a parallelogram and a triangle

7 Find the area of the parallelogram formed by $\mathbf{a} = (2, 1, -1)$ and $\mathbf{b} = (1, 3, 2)$, using the cross product found earlier ($\mathbf{a} \times \mathbf{b} = (5, -5, 5)$).



8 Triangle PQR has $\vec{PQ} = (3, 0, 4)$ and $\vec{PR} = (1, 2, 0)$. Find the area of the triangle.



Properties and applications of the cross product

9 Explain why $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for any vector $\mathbf{a}$, and why $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$ (the cross product is anti-commutative).

10 Find a unit vector perpendicular to both $\mathbf{a} = (1, 2, 0)$ and $\mathbf{b} = (0, 1, 1)$.

The scalar triple product and volume

11 Find the volume of the parallelepiped formed by $\mathbf{a} = (1, 0, 0)$, $\mathbf{b} = (0, 2, 0)$, and $\mathbf{c} = (1, 1, 3)$, using the scalar triple product $V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.

12 Explain what it means, geometrically, if the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ equals zero, and give the name for three vectors satisfying this condition.

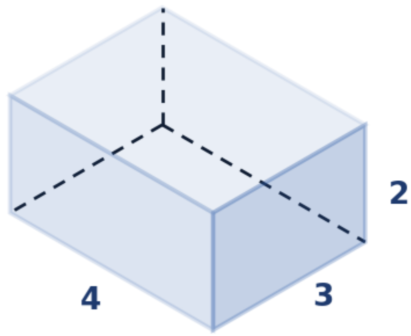
Vector projections

13 Find the scalar projection of $\mathbf{a} = (3, 4, 0)$ onto $\mathbf{b} = (1, 0, 0)$, using $\text{proj} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}$.

14 Find the vector projection of $\mathbf{a} = (2, 3, -1)$ onto $\mathbf{b} = (1, 1, 1)$, using $\text{proj}_{\mathbf{b}}\mathbf{a} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}\mathbf{b}$.

Finding the angle of a parallelepiped's diagonal

15 A rectangular box has edges along the vectors $(4, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 2)$ from one corner. Find the angle between the space diagonal of the box and the edge vector $(4, 0, 0)$, to the nearest degree.



16 Given $\mathbf{a} = (1, 2, 2)$, verify that $|\mathbf{a}| = 3$, and hence find a unit vector in the same direction as $\mathbf{a}$.

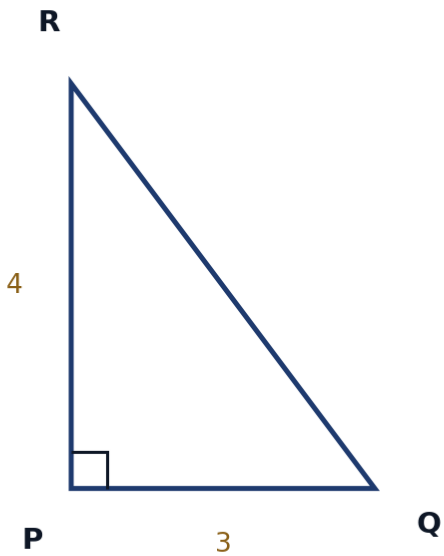
Testing collinearity using vectors

17 Determine whether the points $A(1, 2, 3)$, $B(3, 6, 7)$, and $C(5, 10, 11)$ are collinear, by comparing the vectors $\vec{AB}$ and $\vec{AC}$.

18 Find the value of k such that the vectors $\mathbf{a} = (2, k, -1)$ and $\mathbf{b} = (4, -2, 3)$ are perpendicular.

19 A drone starts at position $(0, 0, 0)$ and flies with constant velocity vector $(3, 4, 2)$ metres per second. Find the drone's position after 5 seconds, and find the distance it has traveled from its starting point (assuming it flies in a straight line).

20 This question has three parts. A triangle has vertices $P(1, 1, 0)$, $Q(4, 1, 0)$, and $R(1, 5, 0)$. (a) Find the vectors $\vec{PQ}$ and $\vec{PR}$. (b) Find $\vec{PQ} \times \vec{PR}$. (c) Hence find the area of triangle PQR, and explain why the cross product points purely in the z-direction here.



21 This question has two parts. Forces $\mathbf{F}_1 = (2, -1, 3)$ N and $\mathbf{F}_2 = (-1, 4, 1)$ N act on an object at the point $(1, 0, 2)$, with position vector $\mathbf{r} = (1, 0, 2)$ measured from the origin. (a) Find the resultant force $\mathbf{F}_1 + \mathbf{F}_2$. (b) Find the moment (torque) of $\mathbf{F}_1$ about the origin, using $\mathbf{M} = \mathbf{r} \times \mathbf{F}_1$.

22 Find the angle that the vector $\mathbf{v} = (1, -2, 2)$ makes with each of the three coordinate axes (its direction angles), to the nearest degree.